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Optimal Trajectory for Active Safe Falls in Humanoid Robots

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Abstract— **Humanoid robots are being introduced in multiple environments, including houses, health care facilities or factories. Bipedal robots are, however, inherently unstable, and they might fall due to multiple reasons, including internal failures or external perturbations. In these situations, the robot should guarantee as much as possible the integrity of humans in the workspace, of the environment, and of the robot itself. When there is some control authority left on the robot, it can be actively commanded to follow a predefined trajectory that minimizes the consequences of the impact with the ground. This paper presents the computation of an optimal falling trajectory using a telescopic inverted pendulum model, which translates into squatting and stretching motions in the robot to dissipate as much energy as possible. The results show that the prescribed trajectory is effective for maximizing the dissipated energy before the actual impact.**

I. Introduction

Humanoid robots are progressively entering new working environments, including homes, offices, factories or even disaster scenarios. Thanks to their bipedal walking, they can for instance step over obstacles and go up and down stairs using robust balancing and walking controllers. As a disadvantage, they are naturally unstable in the upward position and may fall during their locomotion process; this risk is in fact almost inevitable. In controlled environments and labs, the robot is usually connected to a supporting structure to prevent damages in case such a failure occurs. However, in more unstructured environments such external support might not be feasible or desirable, and an unintended fall can lead even to permanent damages in the robot, as displayed for instance during the DARPA Robotic Challenge (DRC) [1].

External perturbations can also make the robot fall. However, several control strategies have been designed to recover stability, in the so-called push-recovery problem. Techniques are in general inspired by natural human reactions such as resisting, leaning or rapidly moving the arms to generate an angular momentum able to recover stability [2]. When the own motions are not enough to recover balance, the robot usually takes a step to a suitable location that prevents the fall, using for instance the concept of capture point [3]. However, with large perturbations also the push recovery or the stepping technique can fail, and the fall will be unavoidable.

When the robot falls, it is desirable to minimize the consequences for humans, for the robot itself and for the surrounding environment, protecting their integrity as much as possible, i.e. calling for a "safe-fall". Depending on the controllability of the robot once it starts to fall down, the falling can be considered as passive or active. In the passive case, the robot very likely fails due to an internal problem (e.g. excessive current of the motors, thus activating the joint breaks), and the robot freezes in one arbitrary pose and falls rigidly to the floor. The only possibility to minimize damages in this case is to use additional protection mechanisms such as airbags [4], [5], [6]. In the active fall there is still some control authority left on the robot once it starts to fall down, therefore the robot can be commanded to follow a sequence of motions that aim to reduce the impact with the ground [7], [8], [9]. This paper is focused on planning an optimal trajectory for minimizing the consequences of such impact with the ground, together with the control logic required for following the desired trajectory during an active fall.

Generally speaking, the controlled fall of a humanoid can be subdivided into three stages:

- Fall Detection: aims to decide the moment where the fall is unavoidable, and marks the switching time from the balancing to the falling controller;
- Controlled Fall: uses a controller that executes all the decisions taken to minimize the damage and/or to modify the falling direction;
- Fall Recovery: consists of a sequence of actions to recover the standing posture.

The design of a fall trigger (FT) able to detect the onset of the fall is not trivial, but is generally based on the trunk attitude and its derivative in the lateral and sagittal planes, collecting and using the information provided by an IMU. Additionally, the horizontal position of the Zero Moment Point (ZMP), together with the position and velocity of the Center of Mass (CoM) can be used in order to give also an estimation of the fall direction. Setting an appropriate threshold for deciding the onset of the fall is difficult, as it seems to be strictly dependent on the task that the robot is accomplishing [8]. For instance, running or jumping are two tasks that require the robot to go through several unstable positions, and the FT could activate the fall controller even if it is not required. For this paper, we will assume that the FT has been already enabled and, consequently, also the fall controller. The recovery of the standing posture is also out of the scope of this paper.

Typical approaches to control the posture of the falling humanoid use energetic considerations, since the impact force is proportional to the total mechanical energy before the impact. A straightforward method is to implement a

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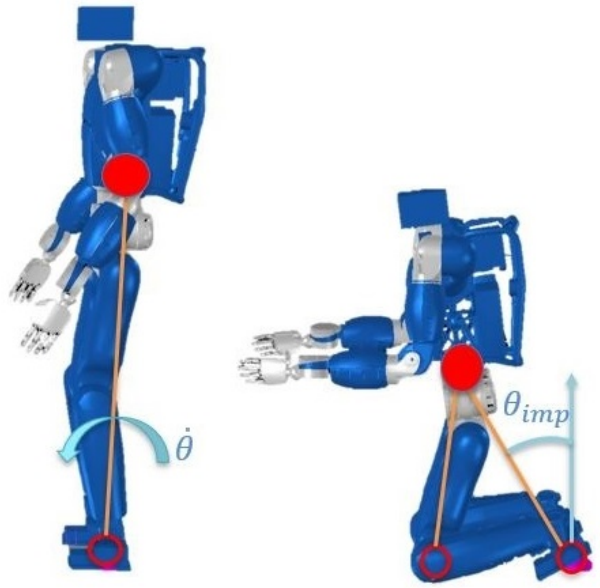


Fig. 1. Humanoid robot modeled as a succession of TIP models for the forward falling direction. Left: standing position; Right: Impact of the first contact point (knee), which becomes the hinge for the next TIP.

squatting motion in order to decrease the potential energy before the collision with the ground [10]. The robot can be modeled as a Telescopic Inverted Pendulum (TIP), and with a suitable heuristic its impact velocity is minimized, therefore reducing the impact force, as presented in [11]. This work was later extended by considering a four-link inverted pendulum to optimize the fall of the robot in the forward direction [12], although the model required a large number of tunable parameters to find a local optimal solution. The TIP model can be also used to generate an online controller that regulates the CoM position in order to dissipate energy before the impact [13]. Prescribed straight [14] or curvilinear trajectories [15] for the CoM can be used and mapped to the multi-link robot, thus generating a squatting and a successive stretching phase in order to dissipate energy.

The optimal selection of successive impact points can be formulated as a Markov Decision Process [16]. These points are later used to model the fall as a succession of simple 2D inverted pendulums, and to generate an offline optimal control for the fall. In case the environment offers possible contact locations and that they are known before the fall, the robot can detect and generate optimal trajectories for the arms such that these additional contact locations can be used to decelerate the robot [9]. An active compliance control on the arms, together with a posture reshape, can also maximize the benefits of a controlled fall [17]. In backward falls, the backpack of the robot can be intentionally used to absorb the impact [18].

This paper is focused on the generation of an optimal trajectory for the CoM of the tipping humanoid using a TIP model that only depends on the physical parameters of the robot and on the initial conditions measured after the fall detection, thus avoiding manual tuning of additional control parameters. The optimality criterion is the minimization of energy at successive impacts, which are modeled as a sequence of inverted pendulums (Fig. 1). Section II describes the proposed method for planning the CoM trajectory, including an analysis of the optimal number of changes in direction for the TIP radial motion during the fall. Section III provides the controller required to follow the prescribed trajectory, and summarizes the online energy-shaping fall control from [13]. That controller is used to compare the benefits of the proposed approach, as discussed in Section IV for forward and backward falls. Section V concludes the paper.

II. Optimal falling trajectory

This section is focused on the problem of generating an optimal CoM trajectory for the falling humanoid, using a TIP model to analyze the mechanical energy of the robot. The whole falling process is modeled as successive TIP models, and the optimal trajectory for these pendula is also analyzed.

A. Impact Force vs Energy Dissipation

The impact force F_{imp} and the total mechanical energy are related through the work-energy principle as $F_{imp} = \Delta E/s$, with s being the deformation distance, and $\Delta E = E^{(-)} - E^{(+)}$ being the difference between the total mechanical energy before and after the impact. In the case of a perfectly inelastic collision the kinetic energy after the impact is zero, $E^{(+)} = 0$, and the impact force will only depend on the total mechanical energy before the impact. Therefore, a decrease in the impact force can be accomplished by decreasing the mechanical energy before the occurrence of the impact, which can be obtained by regulating the CoM position. Additionally, the energetic jump ΔE is entirely dominated by the kinetic energy (as potential energy remains constant), so a minimization of the impact force can be obtained by minimizing the impact velocity of the falling robot.

During the overall robot fall, the initial potential energy due to the CoM height above the ground is transformed into kinetic energy. This kinetic energy must be dissipated at the impact, for instance by distributing the impact over multiple contact points, similar to judo strategies that minimize damages after a competitor falls on the ring [10]. This distribution of the impact also helps to design additional mechanisms at strategic points on the robot that can be prepared to absorb the maximum amount of energy possible, thus reducing the damage on the robot structure. In a forward fall, the robot can use the knees first to break the fall, as they are closer to the ground. Then, a secondary impact point must receive the remaining energy; it can be located at the hands or elbows, which can be relocated to a convenient posture thanks to the high mobility of the arms, thus protecting the neck and head in case that they are also critical parts of the robot.

For the backward case, knees are usually not a feasible contact point. Even in a robot such as the DLR humanoid robot TORO [19], which can bend the knees in the backward direction, a collision on this area is not desirable due to the presence of a parallel bar mechanism that could be seriously damaged. Therefore, the first impact can happen at the lower back, using the elbows as secondary contact point. The elbows could even be directly used as primary contact point. Again, in the case of TORO, for instance, a collision at the lower back or the backpack is not desirable due to the presence of the onboard computers in this area; therefore, a first contact point at the elbows is preferred.

B. Modeling of the fall

The falling humanoid has been modeled using a succession of 2D TIP models with the hinge and the CoM of each pendulum respectively representing the CoP (Center of pressure) and CoM of the robot. Every time a contact point hits the ground at a specific angle θ_{imp}, it becomes the hinge of the next pendulum, and so on until the the end of the fall (Fig. 1). The minimization of the final velocity of each pendulum is required to minimize the impact force at each contact point. This is achieved by designing an optimal trajectory, where the cost function is defined in general terms as:

$$C(\mathbf{x}(t), \mathbf{u}(t), t) = \varphi(\mathbf{x}(t_f), t_f) + \int_{t_0}^{t_f} L(\mathbf{x}(t), u(t), t)dt \quad (1)$$

with $\mathbf{x}(t) = \begin{bmatrix}\dot{r} & r & \dot{\theta} & \theta\end{bmatrix}^T$ being the state vector where $r \in \mathbb{R}$ is the length of the pendulum (distance between the CoP and the CoM of the robot) and θ its tilting angle, and $u(t)$ being a radial control force able to increase or decrease the length of the pendulum. The term φ is known as a *Mayer problem* that defines a penalty term on the state at the final time only, while L is known as a *Lagrange problem* that sets an integral penalty term both on the state and the control function during the whole simulation [20, 184-186]. In order to minimize the final velocity of each pendulum, a penalty term on the velocity at the final time is required, as:

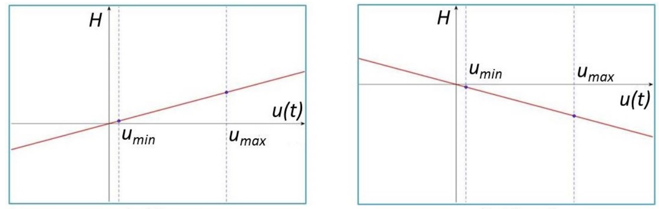
$$\varphi(\mathbf{x}(t_f), t_f) = \left(\sqrt{(\dot{\theta}r)^2 + \dot{r}^2}\right)_{t_f} \quad (2)$$

The cost function is subject to the dynamic constraints defined by the equations of motion. In the state-space form, these equations are:

$$\begin{aligned}
\ddot{r}(t) &= f_1(\mathbf{x}(t), u(t)) = \frac{u}{m} + r\dot{\theta}^2 - g\cos\theta \\
\dot{r}(t) &= f_2(\mathbf{x}(t), u(t)) = \dot{r} \\
\ddot{\theta}(t) &= f_3(\mathbf{x}(t), u(t)) = \frac{g}{r}\sin\theta - \frac{2\dot{r}}{r}\dot{\theta} \\
\dot{\theta}(t) &= f_4(\mathbf{x}(t), u(t)) = \dot{\theta}
\end{aligned} \quad (3)$$

with $m \in \mathbb{R}$ being the total mass of the robot and g the acceleration of gravity. For inserting (3) into (1), we can create the following augmented cost function using Lagrange multipliers:

$$C(\mathbf{x}(t), \mathbf{u}(t), t) = \varphi(\mathbf{x}(t_f), t_f) + \int_{t_0}^{t_f} \left[\lambda^T(\mathbf{f}(\mathbf{x}(t), u(t) - \dot{\mathbf{x}})\right]dt \quad (4)$$



$\lambda_1(t) > 0$ $\lambda_1(t) < 0$

Fig. 2. Graphical representation of the linear relationship between H and $u(t)$. Left: $\lambda_1(t) > 0$; Right: $\lambda_1(t) < 0$

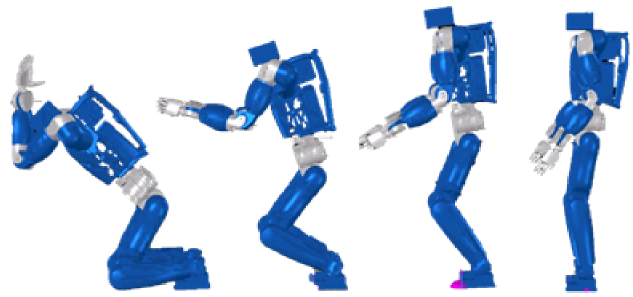


Fig. 3. Typical poses of the humanoid during the primary impact.

This can be simplified by solving the term $\lambda^T\dot{\mathbf{x}}$ by parts, obtaining:

$$C(\mathbf{x}(t), \mathbf{u}(t), t) = \varphi(\mathbf{x}(t_f), t_f) + \int_{t_0}^{t_f} [\lambda^T\mathbf{f}(\mathbf{x}(t), u(t)) + \dot{\lambda}^T\mathbf{x}]dt + \lambda^T(t_0)\mathbf{x}(t_0) - \lambda^T(t_f)\mathbf{x}(t_f) \quad (5)$$

The first term in the integral is the Hamiltonian function H, which can be demonstrated to be linear with respect to $u(t)$ (Fig. 2). The optimal control law $u_{opt}(t)$ will be the one that minimizes $H(\mathbf{x}(t), u(t))$ at each instant of time. Observing Fig. 2, the minimum will always happen at the minimum or maximum value of $u(t)$, depending on the sign of the first adjoint function $\lambda_1(t)$. For this reason, the optimal control law for solving the optimization will be a bang-bang control, with $u(t)$ switching between its minimum and maximum value depending on $\lambda_1(t)$, which will be called hereafter a switching function. However, the adjoint functions are not easy to solve as the impact conditions are not known a priori. For this reason, the optimization should be solved using a different approach.

C. Optimal solution

In order to properly design the bang-bang control for finding the optimal trajectory, the boundaries on the control action $u(t)$ are required. The minimum value is set to be equal to zero: a lower value of the control force (i.e., a force directed from the CoM to the hinge of the pendulum) will accelerate the descending motion of the mass over the gravity acceleration g causing, in a real implementation, the detachment of the feet from the ground. The maximum force exerted on the CoM by all the joints will depend on the pose of the robot since $\mathbf{F}_{max,CoM} = -\mathbf{J}_{CoM}(\mathbf{q})^{-T}\mathbf{F}_{q,max}$,

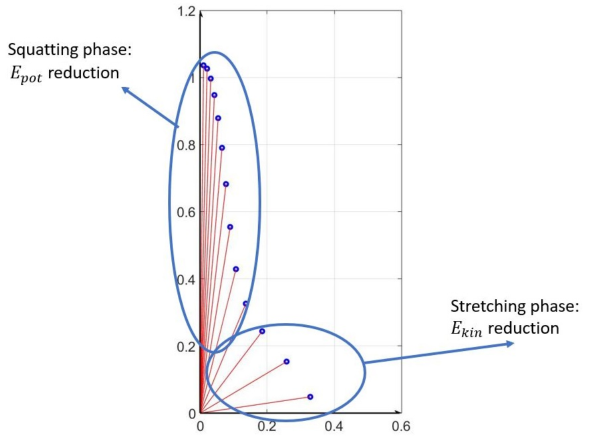


Fig. 4. Typical motion of the TIP model representing the falling humanoid, including a squatting phase and a stretching phase, which respectively aim to decrease E_{pot} and E_{kin}.

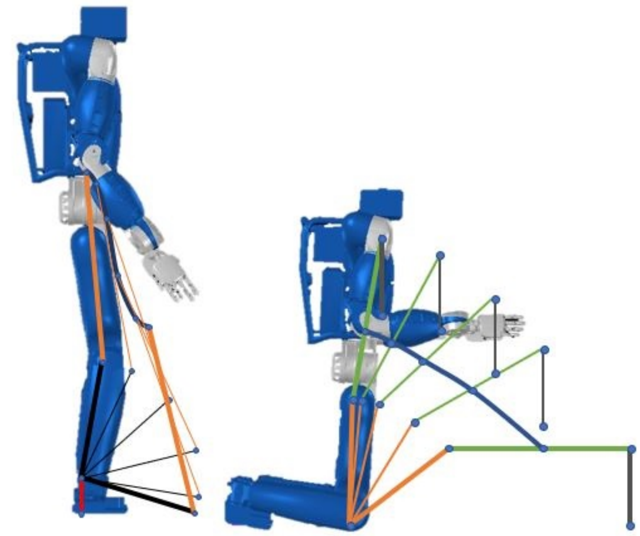


Fig. 5. CoM trajectories generated when falling in the forward direction for the primary and secondary impacts. The optimal CoM trajectory is drawn in blue, and the stick figures indicate intermediate poses of the robot.

with $\mathbf{q}$ being the vector collecting the joints' displacement, $\mathbf{J}_{CoM}$ the CoM Jacobian, and $\mathbf{F}_{q,max}$ the vector containing the maximum torque that can be generated by each joint. For a practical estimation, the robot was frozen in several typical poses during the phases of primary and secondary impact (Fig.3), extracting the CoM Jacobian and calculating $\mathbf{F}_{CoM,max}$ for each of these poses. Then, all the values were averaged, obtaining $u_{max} = 2mg$ for the phase of primary impact, and $u_{max} = 0.8mg$ for the secondary impact.

Due to the differences between the simple TIP model and the real robot embodiment, the impact conditions that define the final time for each phase of the fall must be found, in this case using a simple iterative method. A first optimal trajectory for the TIP was generated using $\theta(t_f)^{(1)} = \pi/2$ as final angle. Using inverse kinematics on the real robot embodiment, the impact angle $\theta_{imp}^{(1)}$ for the first impact with the ground was found. The angle $\theta(t_f)^{(2)}$ for the next iteration is then obtained subtracting half of the difference between $\theta_{imp}^{(1)}$ and $\theta(t_f)^{(1)}$. The iterative method stops when $|\theta(t_f)^{(i)} - \theta_{imp}^{(i)}| < \varepsilon$, with ε being an arbitrary small constant. Naturally, decreasing the value of ε would improve the estimation of the impact angle for each phase, but it also increases the computational time.

The presence of the switching function $\lambda_1(t)$ indicates that the optimal solution for the CoM trajectory during the fall will likely have multiple switches in the control input $u(t)$ for the TIP motion in the radial direction, corresponding to squatting and stretching phases of the humanoid robot. Indeed, the analysis of the falling dynamics of a humanoid robot in previous works has been characterized by a squatting and a successive stretching phase [10], [13], [14], [15], [16], as depicted in Fig.4. The squatting phase aims to reduce the potential energy, which almost entirely defines the total mechanical energy during the first stage of the falling phenomenon. The stretching phase instead reduces the kinetic energy by increasing the mass distribution of the robot around the axis of rotation.

The optimal CoM trajectory should therefore contain squatting and stretching phases, with one or more switches of direction for $u(t)$. To investigate the number of switches required/desired in a falling trajectory, several simulations were run for a robot falling forward with different initial conditions. The falling motion is divided in two stages (Fig. 5): a primary impact on the knees, and a secondary impact on the elbows (the elbows were chosen as secondary point due to the weakness of the hands of TORO). For each case, the velocity (v_{imp}) and the angle of the TIP (θ_{lim}) right before the impact are presented, and the case providing the lowest impact velocity is selected as the preferred case. Table I shows the results obtained for the primary impact, revealing that the optimal number of switches in this phase is equal to one. Table II shows the results for the secondary impact, with an optimal number of switches equal to two for this phase. The optimal number of switches in both falling stages is independent of the magnitude of the perturbation force exerted on the robot.

TABLE I

RESULTS OBTAINED FOR DIFFERENT SWITCHES DURING THE FIRST IMPACT (KNEES)

Switches	Force $[N]$	θ_{lim} $[rad]$	v_{imp} $\left[\frac{m}{s}\right]$
1	1000	0.4713	1.4645
2	1000	0.6622	1.7396
1	1200	0.5214	1.6542
2	1200	0.7529	1.9877
1	1500	0.5791	1.8943
2	1500	0.8289	2.2552

TABLE II

RESULTS OBTAINED FOR DIFFERENT SWITCHES DURING THE SECONDARY IMPACT (ELBOWS)

Switches	Force $[N]$	θ_{lim} $[rad]$	v_{imp} $\left[\frac{m}{s}\right]$
1	1000	1.1300	3.1293
2	1000	1.1664	3.0962
3	1000	1.1497	3.1326
1	1200	1.2459	3.6120
2	1200	1.3241	3.5592
3	1200	1.3213	3.5639
1	1500	1.2818	3.6338
2	1500	1.3120	3.5622
3	1500	1.3036	3.5638

III. CONTROL OF THE FALLING TRAJECTORY

A. Floating base dynamics

A humanoid robot, as a difference to fixed-based robotic manipulators, is not physically attached to the world. For this reason, a floating base description is required in order to set the humanoid position and orientation in the working environment, by adding 6 extra underactuated DoF to the vector of *n* physical joints. The configuration of a humanoid robot is then completely described with respect to the inertial frame as

$$\mathbf{q} = \begin{bmatrix} \mathbf{x}_b^T & \mathbf{q}_r^T \end{bmatrix}^T \tag{6}$$

where $\mathbf{q}_r \in \mathbb{R}^n$ is the joint configuration of the robot with *n* joints, and $\mathbf{x}_b \in \mathbb{R}^6$ is the position and orientation of the floating base. The orientation of the base can be parametrized for instance by using Euler angles. The main problem when dealing with floating base robots is decoupling the task operational space of physical joints from the underactuated floating base. In fact, computing joint velocities via a traditional Jacobian pseudo-inverse method will generate velocities for the 6 DoF of the floating base that cannot be realized due to their underactuated nature. In order to solve this issue, it is mandatory to set some constraints that must be verified during the whole simulation. In particular, it was decided to follow the optimal trajectory $\mathbf{x}_{CoM}$ for the CoM while constraining the feet to be in stationary contact with the ground [21]. Since in the forward case the primary impact occurs at the knees, the constraint is then updated whenever the feet detach from the ground, obtaining:

$$\dot{\mathbf{q}}_{T_1} = \begin{bmatrix} \mathbf{J}_c \\ \mathbf{J}_{CoM} \end{bmatrix}^+ \begin{bmatrix} \mathbf{0} \\ \dot{\mathbf{x}}_{CoM} \end{bmatrix} = \mathbf{J}_{T_1}^+ \dot{\mathbf{x}}_{T_1} \tag{7}$$

with $\mathbf{J}_c \in \mathbb{R}^{13\times n}$ being the constraint Jacobian that includes the two legs (6×2) and the z-axis orientation of the torso, while $\mathbf{J}_{CoM} \in \mathbb{R}^{3\times n}$ is the CoM Jacobian. The pseudo-inverse of $\mathbf{J}_{T_1}$ is denoted by $\mathbf{J}_{T_1}^+$.

While the CoM trajectory found with the method in Section II is followed, the arms are also controlled in order to approach the ground perpendicularly, in order to minimize the resulting forces and torques at the shoulders. However, giving more than one task to the robot could easily lead to kinematic incompatibilities that will generate an unfeasible motion. For this reason, a hierarchical control is required in order to accomplish both tasks T_1 and T_2, by successively projecting onto the nullspace of the previous task. The nullspace N of the first task can be expressed as $\mathbf{N} = \mathbf{I}_n - \mathbf{J}_{T_1}^+ \mathbf{J}_{T_1}$ [22]. Thus, it is possible to declare the second task as

$$\dot{\mathbf{q}}_{T_2} = \mathbf{N} \begin{bmatrix} \mathbf{J}_{re} \\ \mathbf{J}_{le} \end{bmatrix}^+ \begin{bmatrix} \dot{\mathbf{x}}_{re} \\ \dot{\mathbf{x}}_{le} \end{bmatrix} \tag{8}$$

The two velocities for the right and left elbows ($\dot{\mathbf{x}}_{re}$ and $\dot{\mathbf{x}}_{le}$) are calculated using the position error between the actual and desired elbow (end-effector) position $\dot{\mathbf{x}} = \mathbf{K}(\mathbf{x} - \mathbf{x}_{des})$. The desired elbow position is the one that guarantees perpendicularity between arms and ground at the impact time, and $\mathbf{K}$ is a constant gain diagonal matrix.

B. Controller formulation

Once $\dot{\mathbf{q}}_{T_1}$ and $\dot{\mathbf{q}}_{T_2}$ are defined, the desired joint velocities can be calculated as

$$\dot{\mathbf{q}} = \dot{\mathbf{q}}_{T_1} + \dot{\mathbf{q}}_{T_2} \tag{9}$$

The vector in (9) contains $n + 6$ elements representing the velocities of the physical joints and the velocities of the underactuated DoF of the floating base. If the constraints in (7) are always verified during the whole simulation, this can be simplified to

$$\dot{\mathbf{q}}_r = \mathbf{S}(\dot{\mathbf{q}}_{T_1} + \dot{\mathbf{q}}_{T_2}) \tag{10}$$

where $\mathbf{S} = \begin{bmatrix} \mathbf{0}_{(n-6)\times 6} & \mathbf{I}_{(n-6)\times(n-6)} \end{bmatrix}$ is a matrix that selects the actuated joints only. The robot can then be controlled in position, numerically integrating the velocity vector calculated in (10).

C. Energy Shaping Controller

To compare the results of the proposed trajectory-following controller using the optimal trajectory for the CoM, we considered the online energy shaping (ES) controller proposed in [13].

The tipping humanoid has been modeled using the TIP model with the y-axis directed downward, as shown in Fig. 6. The equations of motion are:

$$m\ddot{r} - mr\dot{\theta}^2 + mg\cos\theta = 0 \tag{11}$$

$$mr^2\ddot{\theta} - mgr\sin\theta + 2mr\dot{r}\dot{\theta} = 0 \tag{12}$$

The only parameter that can be controlled during the fall is the length r of the pendulum, which regulates the CoM position, i.e. the pendulum can contract or elongate in order to dissipate energy. The total mechanical energy is given by

$$E = E_K + E_P = \frac{1}{2}mr^2\dot{\theta}^2 + \frac{1}{2}m\dot{r}^2 - mgr\cos\theta \tag{13}$$

Taking the time derivative of (13), substituting $\ddot{\theta}$ from (12) and neglecting the higher order terms (i.e., $\ddot{r}$) [13], the following relation is obtained

$$\dot{E} = 2mgr\dot{\theta}\sin\theta - (mr\dot{\theta}^2 + mg\cos\theta)\dot{r} \tag{14}$$

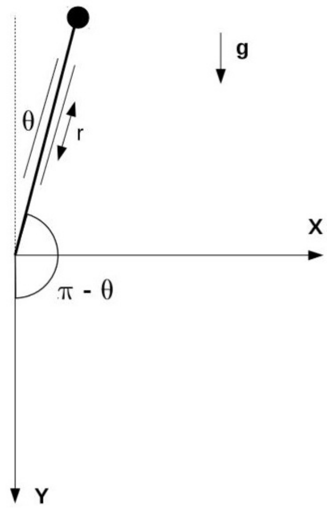


Fig. 6. Telescopic Inverted Pendulum (TIP) model.

Defining $A = 2mgr\dot{\theta}\sin\theta$ and $B = mr\dot{\theta}^2 + mg\cos\theta$, the desired $\dot{r}_d$ is found as

$$\dot{r}_d = K\,B\,E + A/B \tag{15}$$

where $K > 0$ is a control gain. This control logic makes $\dot{E}$ dependent on E as follows

$$\dot{E} = -KB^2E \tag{16}$$

From (16) it is evident that with the chosen input $\dot{r}_d$, the gradient of energy difference will always be decreasing and heading to the desired value $E_d = 0$, which is also the minimum value possible (i.e., the pendulum stops completely before the impact).

IV. Results and discussion

The proposed controller has been tested using a dynamic model of the DLR TORO humanoid robot (27 DoFs, 174 cm tall with a total weight of about 76 kg) [19]. The simulations were run using Matlab and OpenHRP; the robot was pushed in the forward and backward directions with different perturbation forces applied at the torso during $0.05\ s$. To simplify the simulation, the real joint limitation of $\pm45°$ at the ankle was artificially increased to $\pm90°$, such that the feet always remain in contact with the ground during the primary contact phase.

Fig. 7a shows snapshots of the forward fall with a perturbation force of 1200 N, together with the energy trends obtained during the dynamic simulation. As expected, the simulation starts with the robot squatting in order to decrease the potential energy; the robot is free-falling in the first instants of the fall, therefore the kinetic energy increases. This lasts until $\sim 0.35s$ from the beginning of the simulation, when the switch occurs and the radial control force moves from its minimum ($0\ N$) to its maximum ($2mg$). Starting from this moment, the control force opposes the descending motion of the humanoid, dissipating energy before the occurrence of the first impact ($\sim 240\ J$ dissipated, 36.82% of the initial energy). The second phase of the fall dissipates a relatively small amount of energy, and the total mechanical energy is almost constant till the end of the fall.

Similar considerations can be done for the backward fall (using a perturbation force of 700 N), whose results are shown in Fig.7b. In this second case, the tilting robot is modeled using a single TIP model, and the robot directly falls on the elbows. Elongating the time horizon for the single optimization (i.e., the impact on the first contact point occurs later compared with the forward case), the optimal controller is able to dissipate more energy ($\sim 440\ J$ dissipated, 58.46% of the initial energy).

Tables III and IV summarize the results for forward and backward falls using different initial perturbations, respectively. The percentages of reduction are calculated using as reference values the velocities of the uncontrolled free fall of each contact point, $v_{\text{ref}} = \sqrt{v_0^2 + 2gh_0}$.

TABLE III
Velocity reduction for the humanoid falling in the forward direction, two impact points

Force $[N]$	v_{knee} $[m/s]$	Reduction $[\%]$	v_{elb} $[m/s]$	Reduction $[\%]$	E dissipated knee/elbow $[\%]$
1000	0.7612	76.27	3.4159	7.02	39.77 / 3.12
1200	0.8070	74.70	3.5349	3.78	36.82 / 0.75
1500	0.9442	70.40	3.5408	3.62	32.52 / 0.45

TABLE IV
Velocity reduction for the humanoid falling in the backward direction, single impact point

Force $[N]$	v_{elb} $[m/s]$	Reduction $[\%]$	E dissipated $[\%]$
700	2.3648	50.52	58.46
1000	2.6120	45.34	47.25
1200	2.6952	43.60	42.63

The controller that follows the optimal CoM trajectory was also compared with the energy shaping controller of [13] for the same two cases. The optimal controller provides better results in terms of energy minimization before the occurrence of the impact. This is appreciated in Fig. 8a, where the energy trends for the optimal (left) and ES (right) controller are plotted. For the forward fall, the higher amount of energy dissipated with the optimal controller allows a softer impact with the first contact point ($1.3349m/s$ vs. $2.0184m/s$). However, the stronger impact on the knees obtained with the ES controller greatly decelerates the tipping motion of the falling humanoid robot, and a higher energy dissipation can be obtained before the impact with the elbows. In general, the knees are stronger than the elbows and can sustain higher impact forces without any mechanical failure.

When falling in the backward direction, again the optimal controller gives the best results in terms of impact velocity ($2.3865m/s$ vs. $2.4359m/s$) and also in terms of energy dissipation. As a difference to the forward fall, in this case

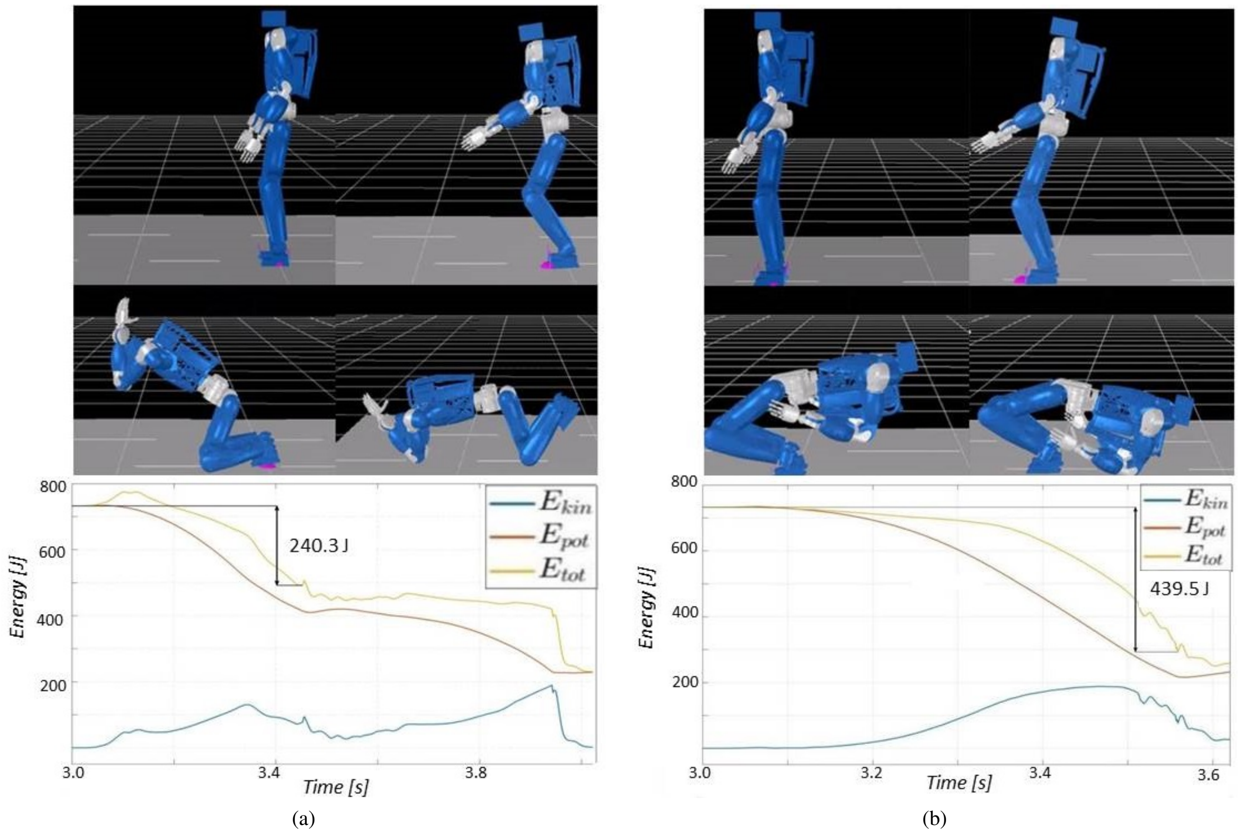


Fig. 7. Results obtained for a) forward and b) backward controlled fall following the optimal CoM trajectory. The evolution of the total mechanical energy (yellow), potential energy (red) and kinetic energy (blue) is displayed, and the energy dissipated before the primary impact is highlighted.

the robot directly falls on the elbows. The optimal controller gives better results since it is able to minimize the impact at this single contact point.

V. Conclusions

This paper presented a solution for computing the optimal CoM trajectory during the active fall of a humanoid robot. The designed optimal controller is able to minimize the impact at each contact point without manually tuning any additional parameters, as a difference to the previous work in [12] that required tuning six different parameters for finding a (locally) optimal trajectory for the CoM. The optimization is performed considering as a cost function the minimization of the impact velocity at the contact points. The impact force was not explicitly computed, but it can be calculated assuming a deformation model [12]. However, as the impact forces expected during the fall were very high [6], the approach was not tested on the humanoid TORO. Experimental implementation of the proposed solution is left as future work.

The formulation of the falling problem can be modified to accommodate additional constraints coming, for instance, from the physical strength of the robot. For example, the maximum impact velocity $v_{imp,des}$ or force that can be mechanically resisted by the knees can be found out (e.g. using a finite element analysis). Then, the cost function expressed in (2) can be modified in order to match this desired impact velocity, thus making sure that there are no structural damages to the robot while exploiting the maximum level of energy dissipation at the primary point. A modified cost function for this case could be expressed as

$$C(\mathbf{x}(t), u(t), t) = v_{imp,des} - \varphi(\mathbf{x}(t_f), t_f) \qquad (17)$$

The results of the proposed optimal controller were also compared with an energy shaping controller previously proposed in [13]. Although the optimal controller provides a trajectory that enables larger dissipation of energy before the impacts, its computational cost is still high, and does not allow at the current state an online implementation. A possible alternative is creating a library of falling trajectories using the optimal solutions proposed in this paper, which would resemble a set of predefined falling “reflexes” for the robot. This library could then be used online to choose the best trajectory for the falling motion given the conditions of the robot and the fall direction once the fall trigger is activated.

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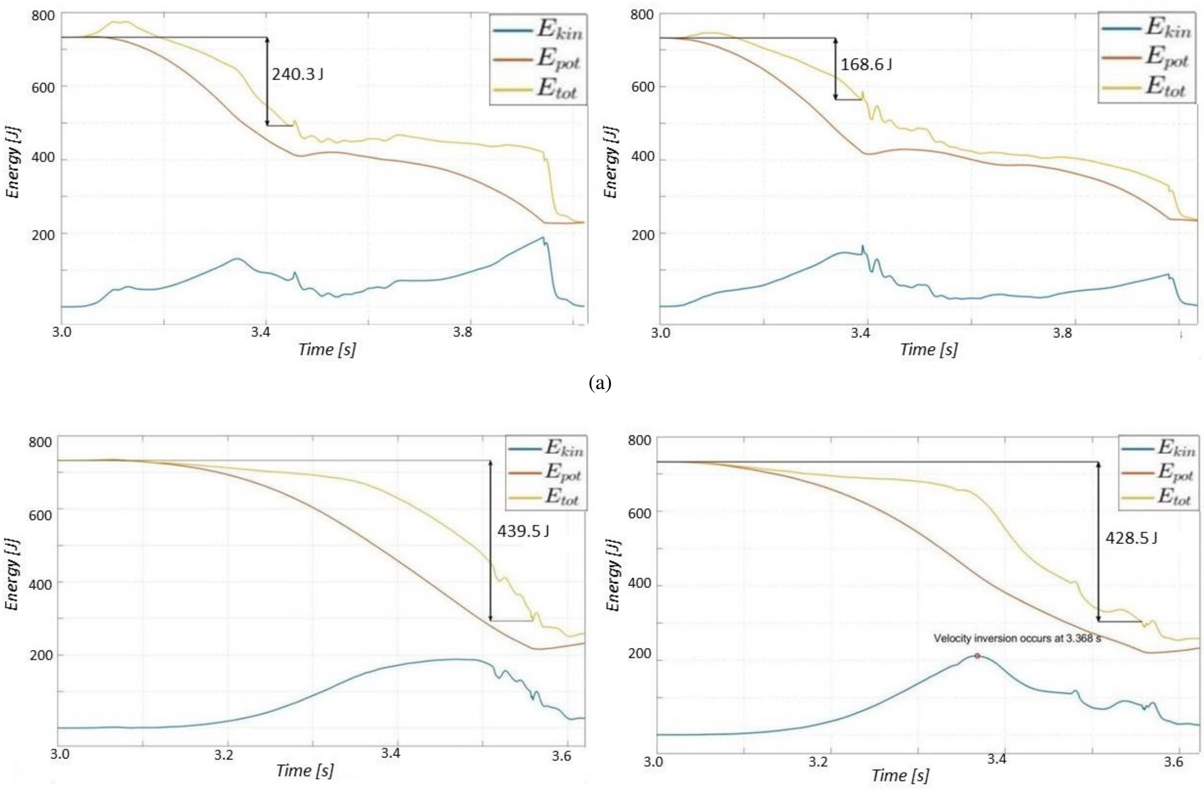

(a)

(b)

Fig. 8. Comparison of the results obtained with the optimal (left) and ES (right) controller. The figure shows the energy trends for the a) forward fall with a perturbation of 1200 N, two contact points; and b) backward falling direction with a perturbation of 700 N, one contact point.

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